VIDYAVARDHAKA COLLEGE OF ENGINEERING MYSURU

Autonomous Institute, Affiliated to VTU, Belagavi

An Internship Report
On
ASIC DESIGN - RTL TO GDS

Submitted in partial fulfilment of the requirement for the award of the degree of
Bachelor of Engineering
In
Electronics and Communication Engineering
By

HARIPRIYA R
4VV21EC046

Internship Carried Out
At
Vivartan Technologies
LLP

Under the Guidance of

Internal Guide
Dr.Chaithanya DJ
Associate Professor
Dept. of ECE
VVCE, Mysuru

External Guide
Chandhan HP
Vivartan Technologies
VVCE, Mysore

Department of Electronics and Communication Engineering

2024-2025

VIDYAVARDHAKA COLLEGE OF ENGINEERING III STAGE, GOKULAM, MYSURU-570002



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

Certified that the internship work entitled **ASIC FLOW RTL TO GDS** carried out by **HARIPRIYA R 4VV21EC046** a bonafide student of Vidyavardhaka College of Engineering in partial fulfilment for the award of Bachelor of Engineering in Electronics and Communication of the Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the said Degree.

NAME & SIGNATURE
OF THE GUIDE

NAME & SIGNATURE
OF THE HOD

EXTERNAL VIVA

NAME OF THE EXAMINERS

SIGNATURE WITH DATE

1.

2.

VIDYAVARDHAKA COLLEGE OF ENGINEERING III
STAGE, GOKULAM, MYSURU-570002

CERTIFICATE

Certified that the internship work entitled **ASIC FLOW RTL TO GDS** carried out by **HARIPRIYA R** ,USN **4VV21EC046** , a bonafide student of Vidyavardhaka College of Engineering in partial fulfilment for the award of Bachelor of Engineering in Electronics and Communication of the Visvesvaraya Technological University, Belgaum during the year 2024-2025. It is certified that, **he/she** has completed the work satisfactorily.

Name & Signature of the Guide

Name & Signature of the Head of Organization

Certificate

VIVARTAN
Learn | Create | Transform

April 27, 2025

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr/Mrs Haripriya R (USN No: 4VV21EC046)** Student of **Vidyavardhaka College of Engineering, Mysuru** has completed Internship Training in our organization from **06.01.2025 to 26.04.2025**

During the period Haripriya R has shown keen interest and completed the Training on ASIC DESIGN: RTL to GDS successfully.

We wish her success in all her endeavours.

For VIVARTAN TECHNOLOGIES LLP.

SUNIL KUMAR H V

DIRECTOR

VIVARTAN TECHNOLOGIES LLP.
LLPIN : AAG-1708
Registered Office : No. 732, 9th Cross, 5th Main, MC Layout Vijayanagar, Bengaluru – 560 040. INDIA
Visit us at : www.vivartan.com **Tel :** +91 988 671 6583 **E-mail :** vivartan.edu@gmail.com

VIVARTAN
Learn | Create | Transform

INTERNSHIP OFFER LETTER

Candidate Name: - **Mr/Ms Haripriya R**

Dear Mr/Ms Haripriya R ,

We are pleased to inform you that you have been selected for an internship position with **Vivartan Technologies LLP**. We are excited to offer you an opportunity to join our team as **Technical Intern.**

As an intern with us, you will have the chance to work alongside experienced professionals, develop your skills, and contribute to exciting projects. Your start date will be **from January 6, 2025.**

To confirm your acceptance of this offer, please sign and return the enclosed copy of this letter.

Welcome to the team!

Best Wishes
Vivartan Technologies



I. DECLARATION

I, Haripriya R, studying in 8th semester of Bachelor of Engineering in Electronics and Communication Engineering at Vidyavardhaka College of Engineering, Mysuru, hereby declare that the presented report of internship carried out at **Vivartan Technologies LLP** which is being submitted by me in the partial fulfilment for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering, from Visvesvaraya Technological University, Belagavi is a record of me carrying out the internship during the academic year 2024-2025, under the guidance of Chandhan HP , Department of Electronics and Communication Engineering, Vidyavardhaka College of Engineering, Mysuru. I further undertake that the matter embodied in the dissertation has not been submitted previously for the award of any degree or diploma by me to any other university or institution.

Date:

Place: Mysuru

II .ABSTRACT

This internship provided comprehensive exposure to digital design and verification methodologies employed in modern VLSI systems. The work primarily focused on understanding and implementing the RTL (Register Transfer Level) design flow, starting from architectural specifications to synthesis. RTL design using hardware description languages like Verilog and SystemVerilog was carried out to model digital circuits with cycle-accurate behavior.

A fundamental component designed during the internship was an ALU (Arithmetic Logic Unit), capable of performing a range of arithmetic and logical operations such as addition, subtraction, AND, OR, and XOR. The ALU design served as a key module for understanding combinational logic, control signals, and RTL-level abstraction. Its verification was carried out using both directed and random testcases to ensure functional correctness across all operations.

Finite State Machine (FSM) design was also a major focus area, involving the modeling of control logic using state diagrams and their corresponding RTL implementations. Both Mealy and Moore machines were explored, offering insights into sequential logic, state transitions, and timing behavior.
Furthermore, the complete ASIC design flow was explored, covering the transition from RTL to gate-level implementation and physical design. Key stages included logic synthesis, static timing analysis (STA), design for test (DFT), place and route (PnR), and physical verification (DRC, LVS), offering a deep insight into the backend stages of chip design.

Additionally, the design and implementation of a FIFO (First In First Out) buffer module were undertaken. Both synchronous and asynchronous FIFO designs were explored, emphasizing proper handling of read/write pointers, status flags (full, empty), and clock domain crossing for robust data transfer. Overall, the internship offered a well-rounded experience in digital design, RTL modeling, FSM development, and ASIC implementation flow, thereby strengthening core competencies in VLSI and preparing for a professional career in semiconductor design and verification.

III. ACKNOWLEDGEMENT

It is our proud privilege to acknowledge the kind of help and guidance received from several people in preparation of this report. It would not have been possible to prepare this report in this form without their valuable help, cooperation and guidance.

First and foremost, we wish to record our sincere gratitude to Management of this college and to our beloved Principal, **Dr. B. Sadashive Gowda**, Vidyavardhaka College of Engineering, Mysuru, for his constant support and encouragement in preparation of this report and for making available library and laboratory facilities needed to prepare this report.

Our sincere thanks to **Dr. Chandrashekar M Patil**, Professor and Head, Department of Electronics and Communication Engineering, VVCE, for his valuable suggestions and guidance throughout the period of Internship.

We express our sincere gratitude to our guide **CHANDHAN HP**, Department of Electronics and Communication Engineering, VVCE, Mysuru for guiding us through Internship. Our numerous discussions with **him/her,** were extremely helpful.

I would like to express my sincere gratitude to **VIVARTAN TECHNOLOGIES LLP, MYSORE** for allowing me to complete the internship work and **Supervisor's Name** for his/her invaluable guidance, support, and encouragement throughout my internship. Their continuous monitoring and the opportunity to learn in a professional environment have greatly enriched my knowledge and experience.

Our sincere thanks to **Dr. Sharath Kumar A J**, Internship Coordinator and **Prof. Kiran Kumar Humse**, Department of Electronics and Communication Engineering, VVCE, Mysuru for having supported the work related to this Internship. Their contributions and technical support in preparing this report are greatly acknowledged.

Place: Mysuru

IV. Table of Contents

Chapter no.	Chapter name	Page no.
I	Declaration	6
II	Abstract	7
III	Acknowledgement	8
IV	Table of Contents	9
V	List of Figures	10
VI	List of Tables	11
Chapter 1	Company Details	12-13
1.1	Company Profile	10
1.2	About the Company	11
1.3	Objectives	11
1.4	Corporate Profile	12
Chapter 2	Introduction to Internship	13-14
2.1	RTL DESIGN	13
2.2	VLSI DESIGN FLOW	13
2.3	FSM	13
2.4	AXI PROTOCOL	14-16
Chapter 3	ASIC FLOW RTL TO GDS	17-18
3.1	Problem Statement	17
3.2	Introduction	18
3.3	Block Diagram	18
Chapter 4	Tasks Performed	19-24
Chapter 5	Learning Outcomes	25
Chapter 6	Conclusion	26

V. List of Figures

Sl no.	Figure	Page no.
Fig.2.1	RTL Design	13
Fig.2.2	VLSI Design Flow	13
Fig.2.2.1	FSM	13
Fig.2.2.2	ASIC Flow	14
Fig.2.3	AXI Protocol	14
Fig.3.1	ASIC Design Flow from RTL to GDSII	18

VI. List of Tables

Sl no.	Figure	Page no.
Table 2.1	Name of table	14
Table 1.3	Objectives	13
Table 3.3.1	Project Flow	18

Chapter-1

Company Details

1.1 Company Profile

Company name:	**Vivartan Technologies LLP.**
Address:	No. 732, 9 th Cross, 5 th Main, MC Layout, Vijayanagar, Bengaluru - 560040
Phone no:	+91 988 671 6583
Email-address:	vivartan.edu@gmail.com
Website:	www.vivartan.com
Class of company:	Private
Founder:	Mr. Sunil Kumar H V
Founded:	September 2009
Company category:	Company limited by shares
Company sub category:	Non-govt company
Corporate services:	Digital Design, Analog and Mixed Signal, Foundation Ips.

Following are the services:

- Train students.
- Connect students and companies.
- Install fundamentals of VLSI
- Develop industry-ready skills.
- Create quality, employable engineering professionals.

Collaborative partner: VVCE Mysore, ATME College of Engineering Mysore, Christ University Bengalore.

1.2 About the Company

We are a group of industry professionals committed to transform learning space in engineering education that creates quality, employable and industry ready engineering professionals. Ideation without Execution is mere Delusion – We started executing our idea of bridging the gap between Industry and Academia in 2009. We see education powerfully confederated with training and development as a tool; to enable people in fulfilling the market demand consistently. Every member of our team is an active working professional from different companies; Passionate in contributing to the student community and to define a profound way of holistic learning resulting in competent talent pool. Our Team comprises of experts from various domains of IC design who are currently working on high end processor designs and cutting-edge technology nodes including the latest need.

1.3 Objectives

- The primary objective of Vivartan Technologies is to bridge the gap between academic education and industry expectations by delivering practical, hands-on training in VLSI and embedded systems. We aim to:
- Transform engineering education through industry-driven learning approaches.
- Empower students with real-world skills and project experience that enhance employability.
- Create a pipeline of industry-ready talent equipped to work on cutting-edge semiconductor technologies.
- Foster holistic development by combining technical knowledge with problem-solving and professional skills.
- Continuously align training programs with evolving industry trends and technological advancements.
- Our goal is to develop competent, confident, and career-ready engineers who can contribute effectively to the global semiconductor ecosystem.

1.4 Corporate Profile

Vivartan Technologies LLP, established in 2009 and headquartered in Mysuru, Karnataka, is a forward-thinking EdTech company dedicated to transforming engineering education by bridging the gap between academia and the semiconductor industry. Founded by a group of experienced industry professionals, Vivartan focuses on developing industry-ready engineers through practical, project-based training in VLSI design, verification, and embedded systems. The company's core services include specialized training programs in RTL design, SystemVerilog, UVM methodology, ASIC flow, and embedded development, along with a comprehensive Postgraduate Diploma in Design Verification. Vivartan also collaborates with academic institutions to integrate hands-on industry content into engineering curricular.

Chapter-2

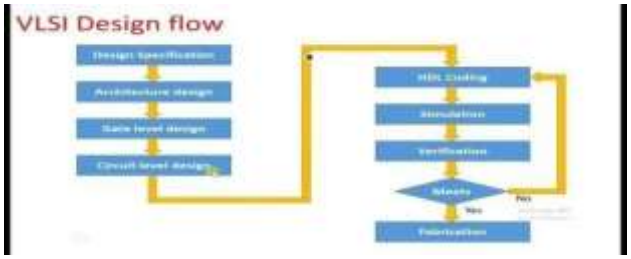
Introduction

2.1 RTL DESIGN



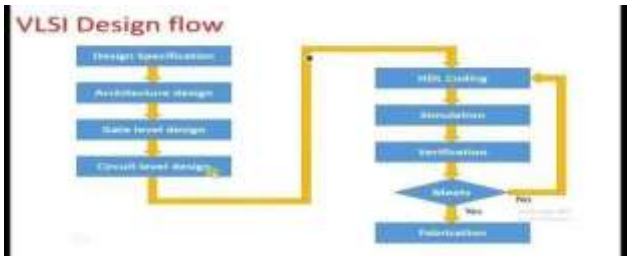
Register Transfer Level (RTL) is an abstraction used in digital circuit design that focuses on the data flow between registers and the operations performed on that data. It represents a system's behavior in terms of registers, the combinational logic that processes data, and the control signals that dictate the operation of the system. At this level, the design is described in hardware description languages (HDLs) such as Verilog or SystemVerilog, which specify the logic in a way that is independent of the physical hardware implementation.

2.2 VLSI DESIGN FLOW



The **VLSI design flow** is a structured, multi-stage process that transforms a high-level specification into a manufacturable chip. It begins with the **specification and requirement analysis**, where the system's functionality, performance, and constraints are defined. The next step is **architectural design**, where the overall structure and key components of the chip are planned. This is followed by **RTL (Register Transfer Level) design**, where the functionality of the chip is described in hardware description languages (HDLs) like Verilog or SystemVerilog.

2.3 VLSI DESIGN FLOW



The **VLSI design flow** is a structured, multi-stage process that transforms a high-level specification into a manufacturable chip. It begins with the **specification and requirement analysis**, where the system's functionality, performance, and constraints are defined. The next step is **architectural design**, where the overall structure and key components of the chip are planned. This is followed by **RTL (Register Transfer Level) design**, where the functionality of the chip is described in hardware description languages (HDLs) like Verilog or SystemVerilog.

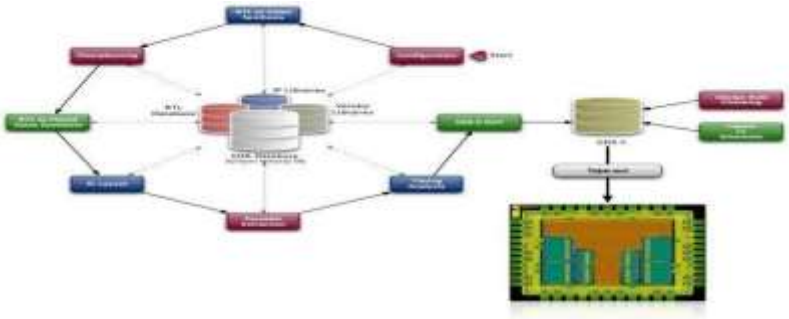
2.3.1 – FSM

A **Finite State Machine (FSM)** is a model used in digital design to describe a system that can be in one of a limited number of states, transitioning between them based on inputs. FSMs are crucial for controlling sequential logic, where the system's behavior depends on both the current state and inputs. There are two main types of FSMs: **Moore machines**, where outputs depend only on the current state, and **Mealy machines**, where outputs depend on both the current state and inputs. FSMs are often used to model control systems, like traffic light controllers or sequence detectors, where the system's operation is dependent on a sequence of inputs.

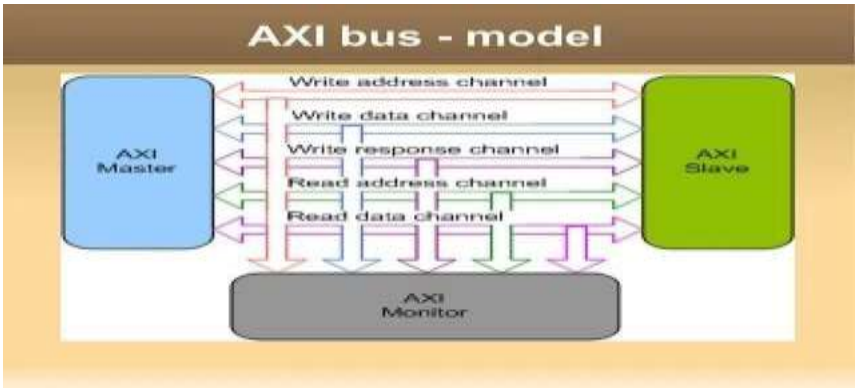


2.3.2 – ASIC FLOW

The **ASIC (Application-Specific Integrated Circuit) design flow** is a structured process that transforms a high-level design into a manufacturable chip. It begins with **specification and requirement analysis**, where the chip's functionality and performance goals are defined. The next step is **architectural design**, where the overall system structure is planned. This is followed by **RTL (Register Transfer Level) design**, where the chip's logic is described using hardware description languages like Verilog or SystemVerilog



2.4 AXI PROTOCOL



AXI (Advanced Extensible Interface) protocol is a high-performance, high-frequency, and low-latency interface standard used for communication between components in System on Chips (SoCs). Part of the ARM AMBA (Advanced Microcontroller Bus Architecture) family, AXI is designed to handle data transfer between master and slave devices in a flexible and efficient manner. It supports multiple features such as burst-based transactions, out-of-order transactions, and low-latency responses. AXI operates in a pipelined manner, which allows multiple transactions to be processed simultaneously, enhancing throughput. It includes separate read and write channels, which improves performance by allowing simultaneous data transfer. The protocol also supports addressing, data transfer, and control signaling. t represents a system's behavior in terms of registers, the combinational logic that processes data, and the control signals that dictate the operation of the system. At this level, the design is described in hardware description languages (HDLs) such as Verilog or SystemVerilog, which specify the logic in a way that is independent of the physical hardware implementation.

Chapter-3

ASIC FLOW RTL TO GDS

3.1 Problem Statement

The design and development of Application-Specific Integrated Circuits (ASICs) involve a complex flow from Register Transfer Level (RTL) to the final GDSII layout. This flow requires seamless integration of multiple design and verification processes to ensure that the chip meets the desired specifications for performance, power consumption, and area.

3.2 Introduction

3.2.1 Overview

The ASIC design flow is a structured and systematic process used to create Application-Specific Integrated Circuits (ASICs), which are custom-designed chips tailored for specific applications. This process spans from the RTL (Register Transfer Level) design stage, where the chip's functionality is described, to the final GDSII file, which contains the physical layout ready for fabrication. The flow involves both frontend and backend design stages, each with its own set of tools, techniques, and checks. Several sign-off checks, such as Static Timing Analysis (STA) and Design Rule Checks (DRC), are conducted before generating the GDSII file. This file is the final representation of the chip's layout and is sent to the semiconductor foundry for tape-out and fabrication. The goal throughout this process is to create a chip that meets all functional specifications while being efficient in terms of performance, power, and area.

3.2.2 Purpose

The **purpose** of the **ASIC design flow** from **RTL to GDSII** is to systematically transform a high-level functional description of a chip into a final, manufacturable design. This process ensures that the chip meets specific application requirements while optimizing performance, power, and area. The flow begins with creating a detailed **Register Transfer Level (RTL)** model that captures the chip's behavior, followed by verification to ensure correctness. Once the design is verified, it moves to synthesis, where the RTL is converted into a gate-level netlist.

3.3 Block Diagram

The ASIC design flow from RTL to GDSII is a comprehensive and systematic process that transforms a high-level digital design into a physical layout ready for fabrication. It begins with the creation of an RTL (Register Transfer Level) model using hardware description languages like Verilog or SystemVerilog, which describes the logical functionality of the chip. This RTL design undergoes functional verification through simulations to ensure it behaves correctly. Once verified, the design moves into the synthesis stage, where the RTL is converted into a gate-level netlist using standard cell libraries.

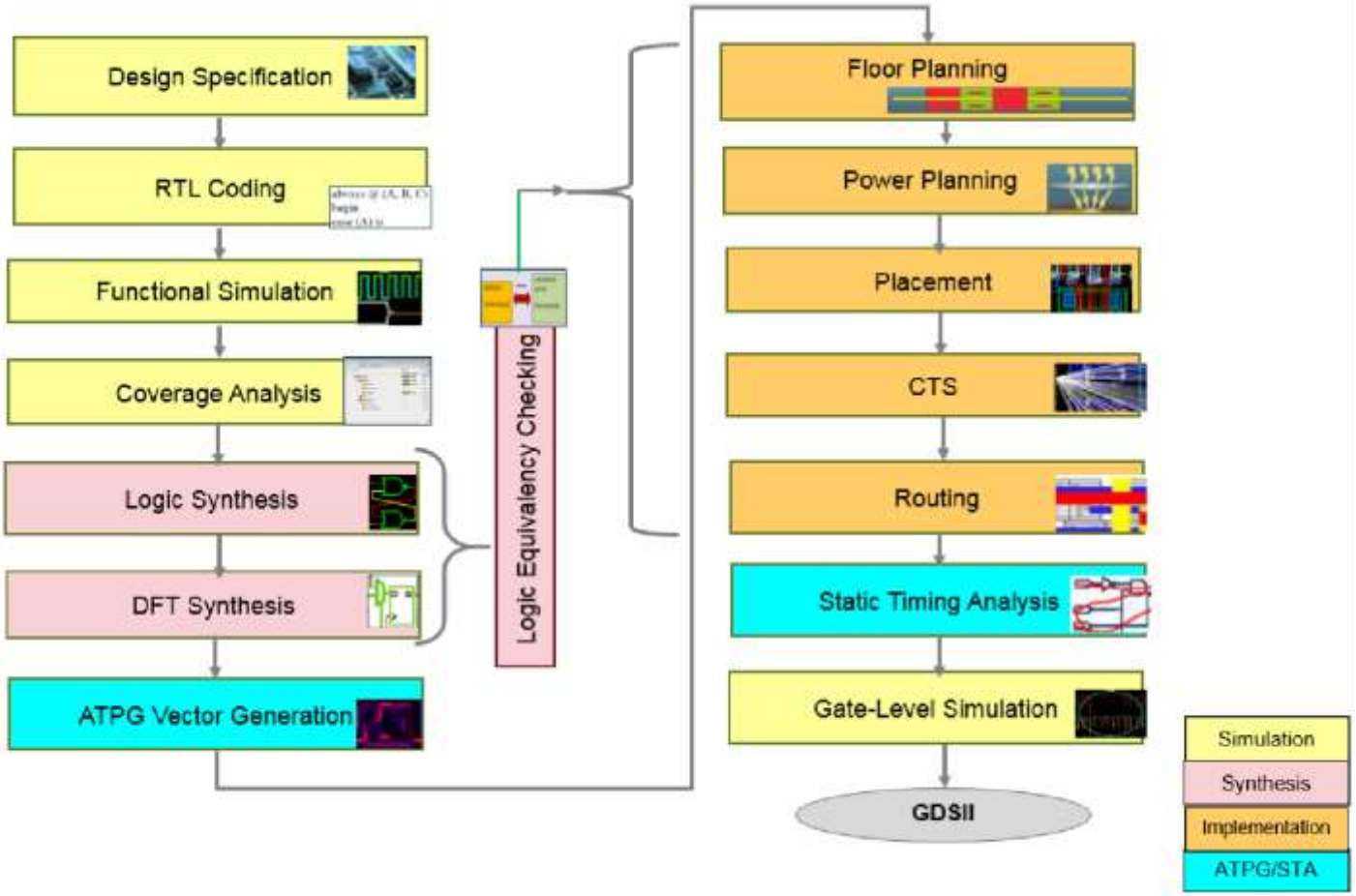


Fig.3.1 ASIC Design Flow: From RTL to GDSII

3.3.1 Project Flow

- RTL Design – Wrote SystemVerilog code to model the design.
- Testbench & Simulation – Verified functionality using simulations.
- Synthesis – Converted RTL to gate-level netlist.
- Floorplanning – Defined layout and positioned major blocks.
- Placement & CTS – Placed cells and built clock tree.
- Routing – Connected all components with metal layers.
- Sign-off Checks – Performed STA, DRC, LVS, and power analysis.
- GDSII Generation – Created final layout file for fabrication.
- Report Preparation – Documented results and submitted final report.

Chapter -4

Task Performed

Task 1: UP-DOWN COUNTER.

Step1. Write code of the design along with test benc Step2. Select appropriate simulator and run options. Step3. RUN Simulation.

Open

```
module updown (
    input clk,
    input rst,
    input up_down,
    output reg [2:0] Q
);

    always @(posedge clk or posedge rst) begin
        if (rst)
            Q <= 3'b000;
        else if (up_down)
            Q <= Q + 1;
        else
            Q <= Q - 1;
    end
endmodule
```

```
module tb_updown;

    reg clk, rst, up_down;
    wire [2:0] Q;

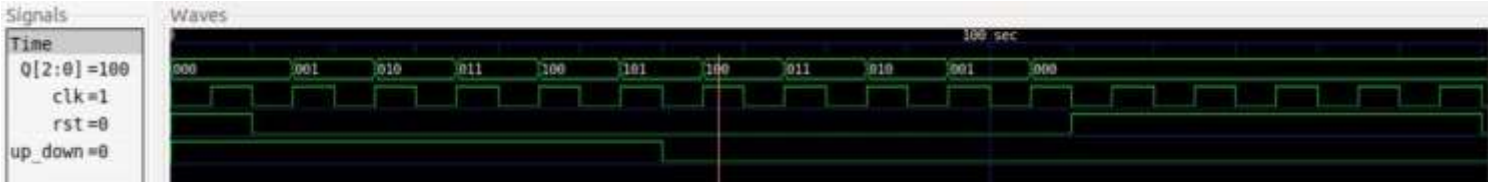
    updown uut (
        .clk(clk),
        .rst(rst),
        .up_down(up_down),
        .Q(Q)
    );

    always #5 clk = ~clk;

    initial begin
        clk = 0; rst = 1; up_down = 1;
        #10 rst = 0;
        #50 up_down = 0;
        #50 rst = 1;
        #50 rst = 0;
        #50 $stop;
    end

    initial begin
        $monitor("Time = %0t | up_down = %b | Q = %b", $time, up_down, Q);
    end

    initial begin
        $dumpfile("updown.vcd");
        $dumpvars(0, tb_updown);
    end
endmodule
```

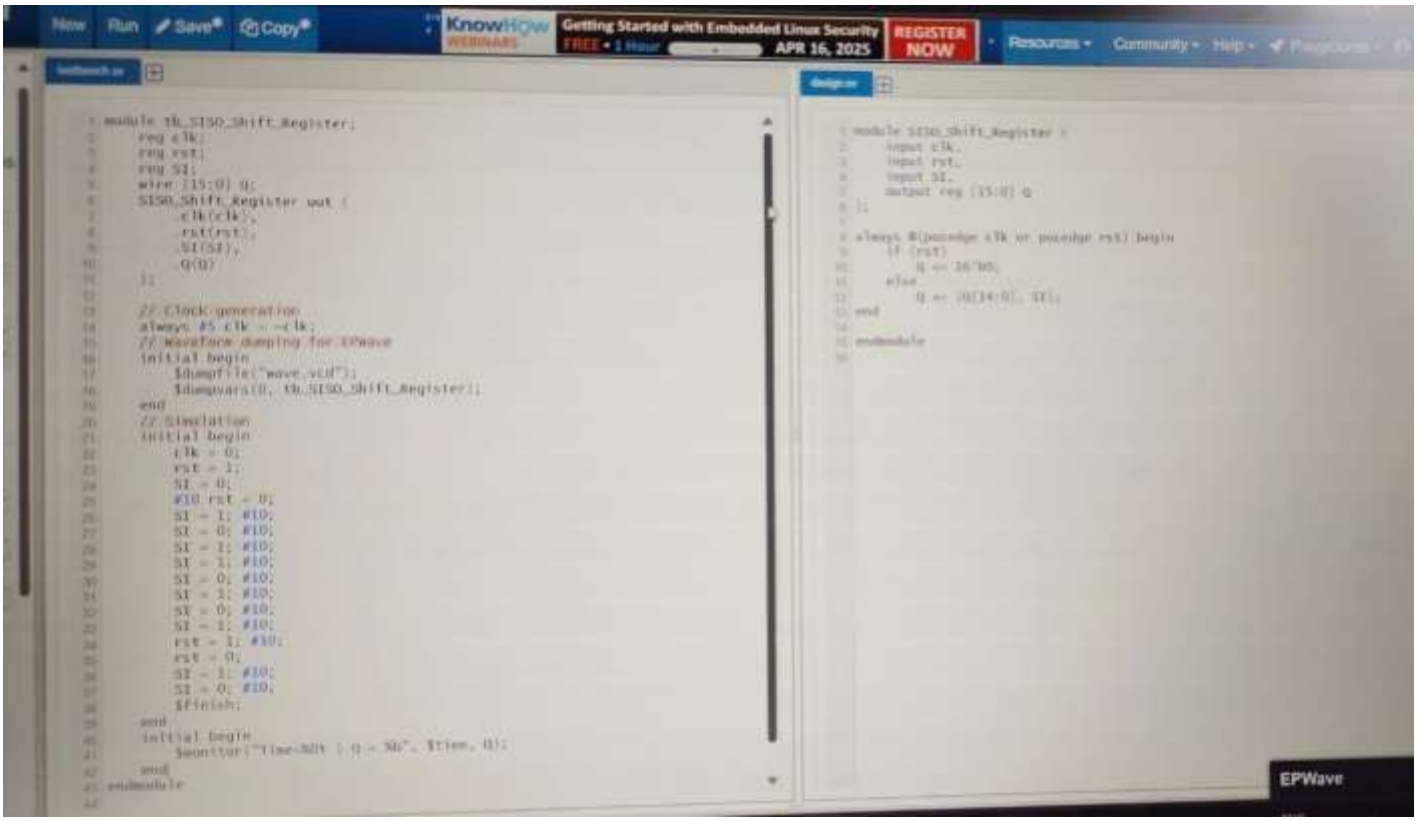


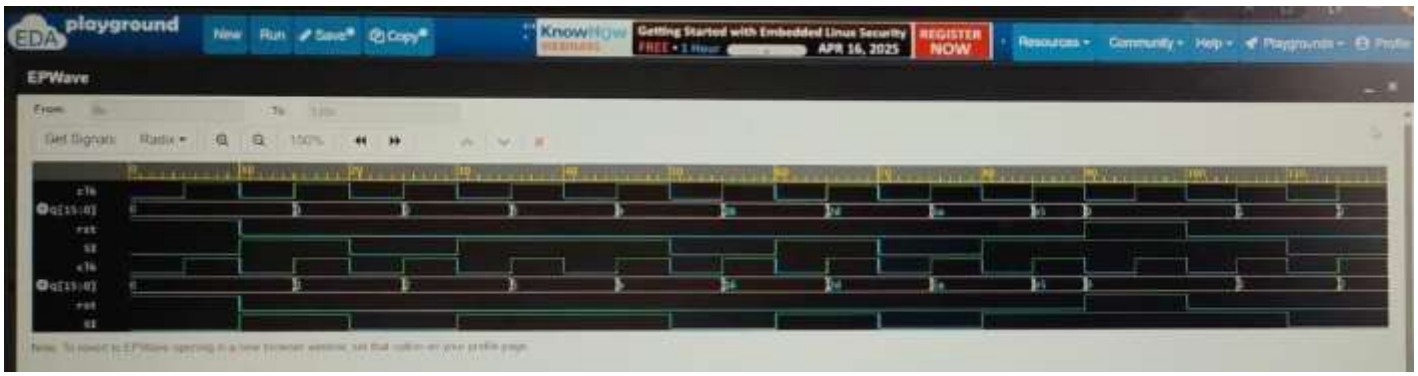
Task2: 16 bit shift register SISO

STEP 1: Design: Create your module and define its behavior.

STEP 2: Verify: Write a testbench to stimulate your design with inputs.

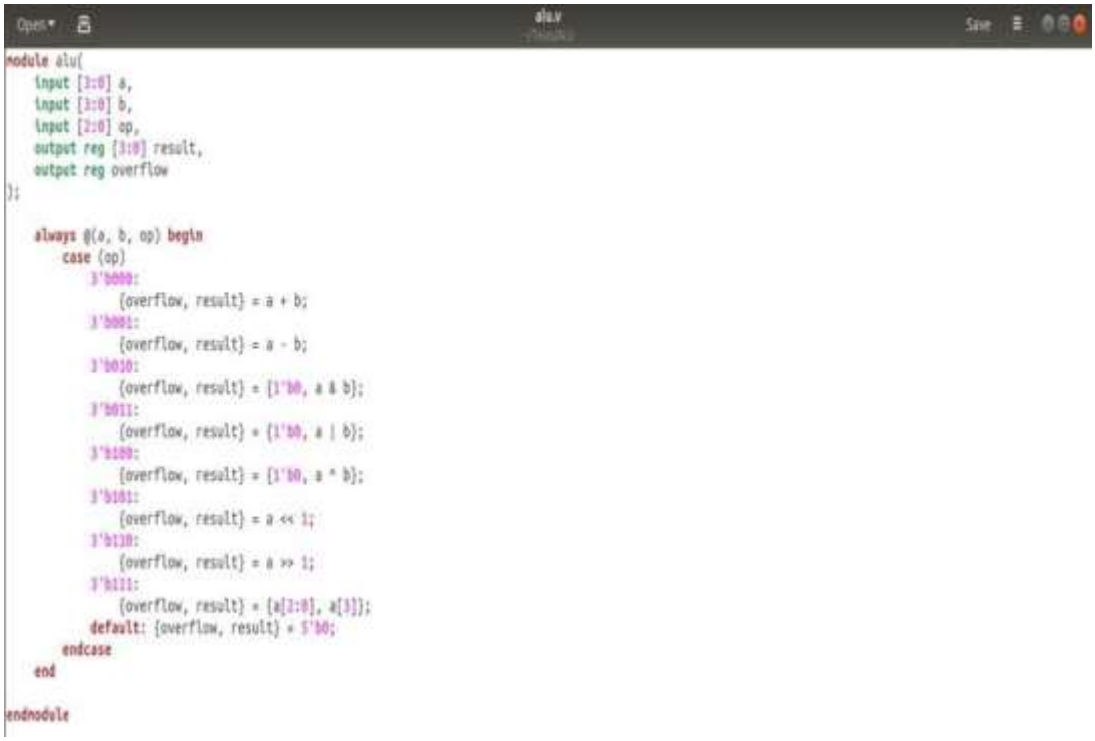
STEP 3: Simulate: Run the simulation, analyze the outputs, and check the waveforms for correctness.





Task 3: ALU Design

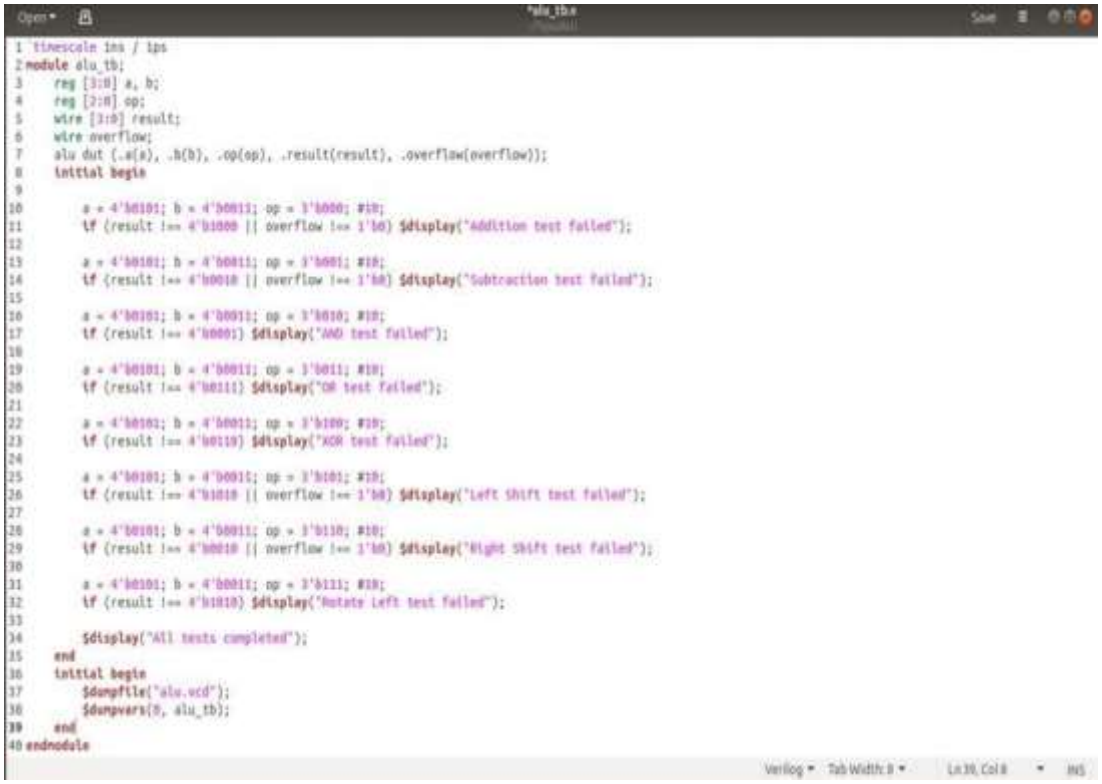
Step1. Write code of the design along with test bench.
Step2. Select appropriate simulator and run options.
Step3. RUN Simulation.



```
module alu(
    input [3:0] a,
    input [3:0] b,
    input [2:0] op,
    output reg [3:0] result,
    output reg overflow
);

    always @(a, b, op) begin
        case (op)
            3'b000:
                {overflow, result} = a + b;
            3'b001:
                {overflow, result} = a - b;
            3'b010:
                {overflow, result} = {1'b0, a & b};
            3'b011:
                {overflow, result} = {1'b0, a | b};
            3'b100:
                {overflow, result} = {1'b0, a ^ b};
            3'b101:
                {overflow, result} = a << 1;
            3'b110:
                {overflow, result} = a >> 1;
            3'b111:
                {overflow, result} = {a[2:0], a[3]};
            default: {overflow, result} = 5'b0;
        endcase
    end

endmodule
```



```
1  timescale 1ns / 1ps
2 module alu_tb;
3     reg [3:0] a, b;
4     reg [2:0] op;
5     wire [3:0] result;
6     wire overflow;
7     alu dut (.a(a), .b(b), .op(op), .result(result), .overflow(overflow));
8     initial begin
9
10        a = 4'b0101; b = 4'b0011; op = 3'b000; #10;
11        if (result !== 4'b1000 || overflow !== 1'b0) $display("Addition test failed");
12
13        a = 4'b0101; b = 4'b0011; op = 3'b001; #10;
14        if (result !== 4'b0010 || overflow !== 1'b0) $display("Subtraction test failed");
15
16        a = 4'b0101; b = 4'b0011; op = 3'b010; #10;
17        if (result !== 4'b0001) $display("AND test failed");
18
19        a = 4'b0101; b = 4'b0011; op = 3'b011; #10;
20        if (result !== 4'b0111) $display("OR test failed");
21
22        a = 4'b0101; b = 4'b0011; op = 3'b100; #10;
23        if (result !== 4'b0110) $display("XOR test failed");
24
25        a = 4'b0101; b = 4'b0011; op = 3'b101; #10;
26        if (result !== 4'b1010 || overflow !== 1'b0) $display("Left Shift test failed");
27
28        a = 4'b0101; b = 4'b0011; op = 3'b110; #10;
29        if (result !== 4'b0010 || overflow !== 1'b0) $display("Right Shift test failed");
30
31        a = 4'b0101; b = 4'b0011; op = 3'b111; #10;
32        if (result !== 4'b1010) $display("Rotate Left test failed");
33
34        $display("All tests completed");
35    end
36    initial begin
37        $dumpfile("alu.vcd");
38        $dumpvars(0, alu_tb);
39    end
40 endmodule
```



Task 4: 32 Bit RAM

Step1. Write code of the design along with test bench.
Step2. Select appropriate simulator and run options.
Step3. RUN Simulation.

```
module ram_32x32_tb;
    reg clk, we;
    reg [4:0] address;
    reg [31:0] data_in;
    wire [31:0] data_out;

    // Instantiate the RAM module
    ram_32x32 RAM (
        .clk(clk),
        .we(we),
        .address(address),
        .data_in(data_in),
        .data_out(data_out)
    );

    // Clock Generation
    initial begin
        clk = 0;
        forever #5 clk = ~clk; // Clock toggles every 5 ns
    end

    initial begin
        $dumpfile("ram_32x32_tb.vcd");
        $dumpvars(0, ram_32x32_tb);
        $monitor("Time=%0t | we=%b | addr=%d | data_in=%h | data_out=%h",
$time, we, address, data_in, data_out);

        // Write Operation
        we = 1;
        address = 5'd0; data_in = 32'hAABBCCDD; #10;
        address = 5'd1; data_in = 32'h12345678; #10;
        address = 5'd2; data_in = 32'hDEADBEEF; #10;

        // Read Operation
        we = 0;
        address = 5'd0; #10;
        address = 5'd1; #10;
        address = 5'd2; #10;

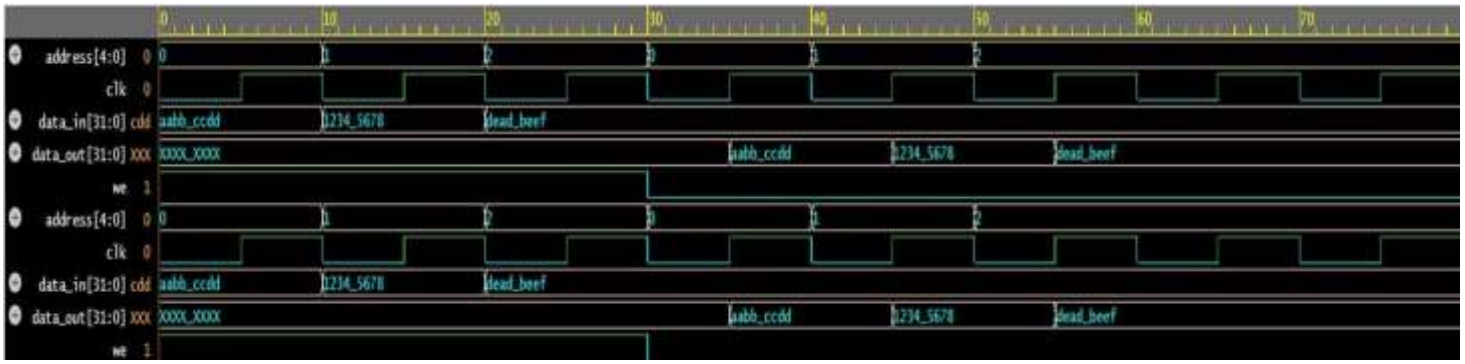
        #20 $finish;
    end
endmodule
```

```
module ram_32x32 (
    input clk,              // Clock signal
    input we,               // Write Enable (1 = Write, 0 = Read)
    input [4:0] address,    // 5-bit address (32 locations)
    input [31:0] data_in,   // 32-bit data input
    output reg [31:0] data_out  // 32-bit data output
);

    // Memory Array: 32 locations of 32-bit each
    reg [31:0] mem [0:31];

    always @(posedge clk) begin
        if (we)
            mem[address] <= data_in;  // Write Operation
        else
            data_out <= mem[address]; // Read Operation
    end

endmodule
```



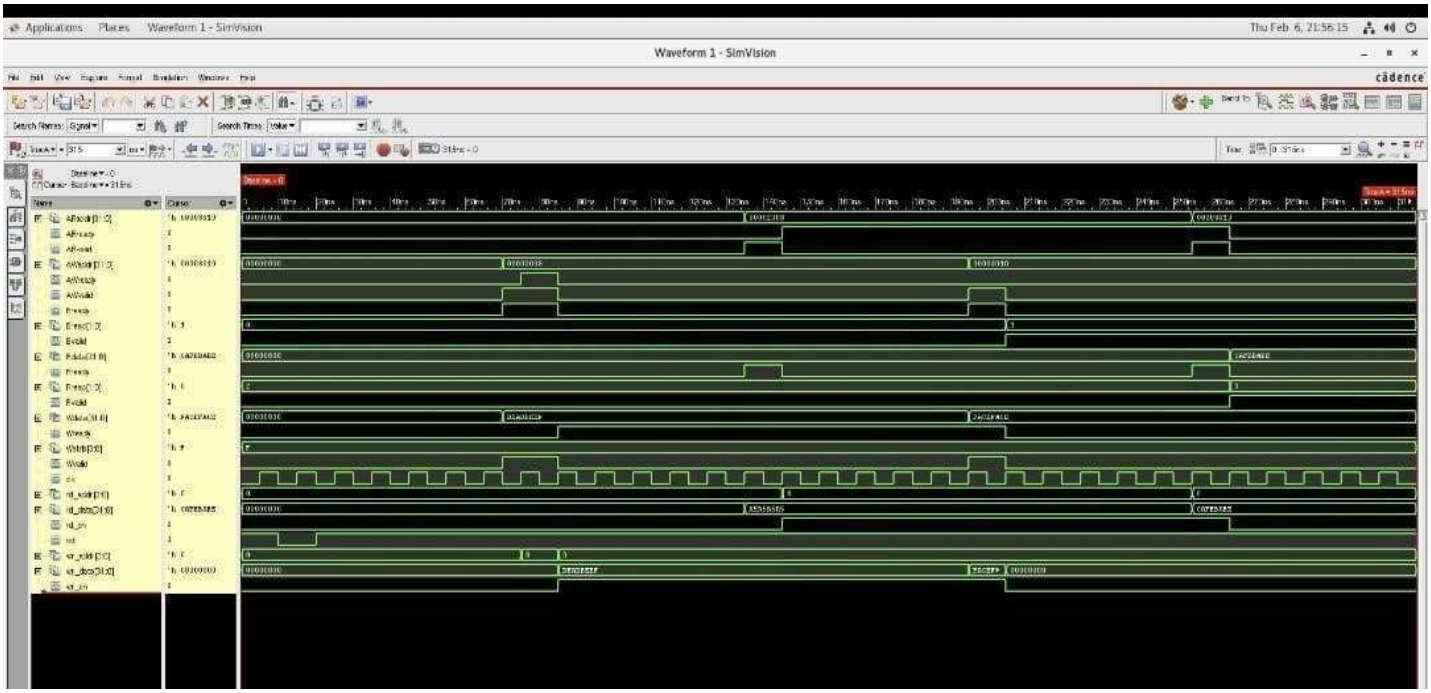
Task 5: AXI

Step1. Write code of the design along with test benc
Step2. Select appropriate simulator and run options.
Step3. RUN Simulation.

OUTPUT






Applications
Places
Waveform 1 - SimVision
Thu Feb 6, 21:56:15
Waveform 1 - SimVision
cadence

Chapter – 5

Learning Outcomes

By studying the ASIC design flow from RTL to GDSII, students will gain a deep understanding of the entire process involved in the creation of custom integrated circuits. Starting from the RTL (Register Transfer Level) design, learners will explore how to write and verify functional descriptions of a digital system using hardware description languages (HDL) like Verilog or VHDL. They will learn to simulate and test their designs using testbenches to ensure that the functionality is correct before moving on to the next stage. The next step involves synthesis, where students will understand how to convert their RTL code into a gate-level netlist using automated tools such as Synopsys Design Compiler. This netlist, which describes the logic gates and interconnections, is then optimized based on the constraints of area, power, and performance.

In the physical design phase, learners will explore the process of placement and routing, where components (gates) are placed on the chip's layout, and the physical connections between them are made. Students will become familiar with concepts like clock tree synthesis, timing analysis (STA), and design rule checking (DRC), which ensure that the design meets both functional and physical constraints. Additionally, they will understand the importance of LVS (Layout vs. Schematic) checks to verify the correctness of the design at the physical level.

Furthermore, students will gain knowledge about the file formats used throughout the flow, such as netlist formats for logic representation, SDF (Standard Delay Format) for timing information, and the GDSII format, which is used for the final layout and is essential for manufacturing the chip. By the end of this course, students will have developed the skills necessary to perform the key tasks involved in ASIC design, while also gaining a strong grasp of the trade-offs between speed, area, and power consumption that designers face at each stage. This experience will prepare them for real-world challenges in VLSI design and semiconductor manufacturing.

Chapter – 6

Conclusion

In conclusion, the ASIC design flow from RTL (Register Transfer Level) to GDSII (Graphic Data System II) is a comprehensive and iterative process that involves transforming abstract functional descriptions of a digital system into a physical chip layout ready for manufacturing. Each step in this flow is critical, from writing the RTL code, which defines the functionality of the circuit, to performing functional verification through simulation and testbenches. The synthesis phase then converts this RTL code into a gate-level netlist, optimizing it for performance, area, and power consumption.

SUSTAINABLE DEVELOPMENT GOALS



Mapping and Justification

The ASIC design flow, mapping and justification play crucial roles in ensuring that the design meets the required specifications, including functionality, performance, power consumption, and physical layout. Mapping refers to the process of converting the high-level RTL (Register Transfer Level) design into a gate-level netlist using logic gates and cells from the technology library. This stage ensures that the logic described in RTL is accurately represented in terms of gates, with optimization for area, speed, and power based on design goals. During synthesis, the mapping process ensures that the design is both functionally correct and optimized for the given constraints.

Once the design is mapped to a gate-level netlist, justification begins, which involves validating that the design satisfies specific performance and physical constraints. This includes timing justification, where static timing analysis (STA) is performed to ensure that the design meets the required setup and hold times for flip-flops. If timing violations occur, designers must adjust placement or select different gates to correct the timing issues. Power justification is also performed to ensure that the design operates within the power budget, often requiring adjustments to the gate selection or the use of low-power techniques like clock gating. Additionally, area justification ensures that the layout fits within the designated chip area, optimizing the placement of gates and minimizing wasted space